財政學 HW3

1. (Hyman 11.3) The price elasticity of demand for wine is estimated to be -1 at all possible quantities. Currently, 200 million gallons of wine are sold per year, and the price averages $6 per bottle. Assuming that the price elasticity of supply of wine is 1 and the current tax rate is $1 per bottle, calculate the current excess burden of the tax on wine. Suppose the tax per bottle is increased to $2 per bottle. What will happen to the excess burden of the tax as a result of the tax increase? Under what circumstances can a doubling of the tax on wine actually improve resource use in the United States, despite the increase in the excess burden of the tax?

ANS

Because both the price elasticity of demand and of supply of wine is unity, the excess burden is

$$1/2(200/6)\times(1/2) = \$8,333,333.33 \text{ per year}$$

If the tax rate goes up to two, the excess burden will quadruple to

$$1/2(4)(200/6)\times(1/2) = \$33,333,333.33$$

If there are negative externalities associated with drinking wine, then the tax can actually improve resource use despite its high excess burden.

2. (Hyman 11.5) Suppose the supply of housing construction is infinitely elastic at a price of $150 per square foot. Currently 1 million square feet are built per month. If the price elasticity of demand for housing is -1, calculate the monthly excess burden of a 10 percent tax on housing construction. (Hint: Go to the discussion on taxation of constant cost industries.) What is the monthly excess burden if the tax is 20 percent? Who will bear the incidence of the tax?

ANS

The excess burden is 0.5(0.1)2×150,000,000(–1) = – $750,000 per month. If the tax rate is doubled to 20 percent, the excess burden quadruples to a loss of $3 million per month. Because the supply is perfectly elastic, the incidence of the tax is borne fully by the buyers. For a 10 percent tax, the market equilibrium price per square foot of construction will rise to $165. For a 20 percent tax, the market equilibrium price will rise to $180.

3. (Rosen 15.1) Which of the following is likely to impose a large excess burden?

a. A tax on land.

b. A tax of 24 percent on the use of cellular phones. (This is the approximate sum of

federal and state tax rates in California, New York, and Florida.)
c. A subsidy for investment in "high-tech" companies.
d. A tax on soda bought in a cup or glass but not bought in a bottle or can. (Such a tax exists in Chicago.)
e. A 10-cent tax on a deck of cards. (Such a tax exists in Alabama.)
f. A tax on blueberries. (Such a tax exists in Maine.)

ANS
a. The supply of land is fixed, or perfectly inelastic, so there is no excess burden because the lower price that sellers receive does not cause quantity supplied to fall.
b. The use of cell phones is probably fairly price-elastic, which implies that the excess burden could be large.
c. It is possible that companies could identify themselves as high-tech in order to receive the subsidy. Thus, the supply is quite elastic, and there will be substantial excess burden.
d. Consumers and sellers will likely agree to avoid cups and glasses in order to avoid the tax. A tax that is easily avoided does not have much of an impact, except to create some inconvenience, and does not raise revenue.
e. Card companies can easily increase or decrease the number of cards in a pack, avoiding the tax and reducing the excess burden.
f. There are many good substitutes for blueberries. Therefore, their demand is quite elastic, and a tax on them will have a substantial excess burden, relative to the size of revenues collected.

4. (Rosen 15.9) In the United Kingdom, each household that owns a television pays a compulsory levy that is equivalent to \$233 per year. The total revenue collected, which is over \$7 billion annually, goes to the British Broadcasting Corporation. Do you think that such a tax is likely to have a substantial excess burden relative to the revenues collected?

ANS
It is likely that the demand for owning at least one television is quite inelastic. It follows that the excess burden from a \$233 per year television tax is small relative to the revenues that are collected.

5. (Rosen 15.12) In an effort to reduce alcohol consumption, the government is considering a \$1 tax on each gallon of liquor sold (the tax is levied on producers). Suppose that the supply curve for liquor is upward sloping and its equation is

Q=30,000P (where Q is the number of gallons of liquor and P is the price per gallon). The demand curve for liquor is Q=500,000-20,000P.

a. Draw a sketch to illustrate the excess burden of the tax. Next use algebra to calculate the excess burden. Show graphically the excess burden generated by the $1 unit tax. (Hint: Compare the losses of both consumer and producer surplus to tax revenues.)

b. Suppose that each gallon of liquor consumed generates a negative external cost of $0.50. How does this affect the excess burden associated with the unit tax on liquor?

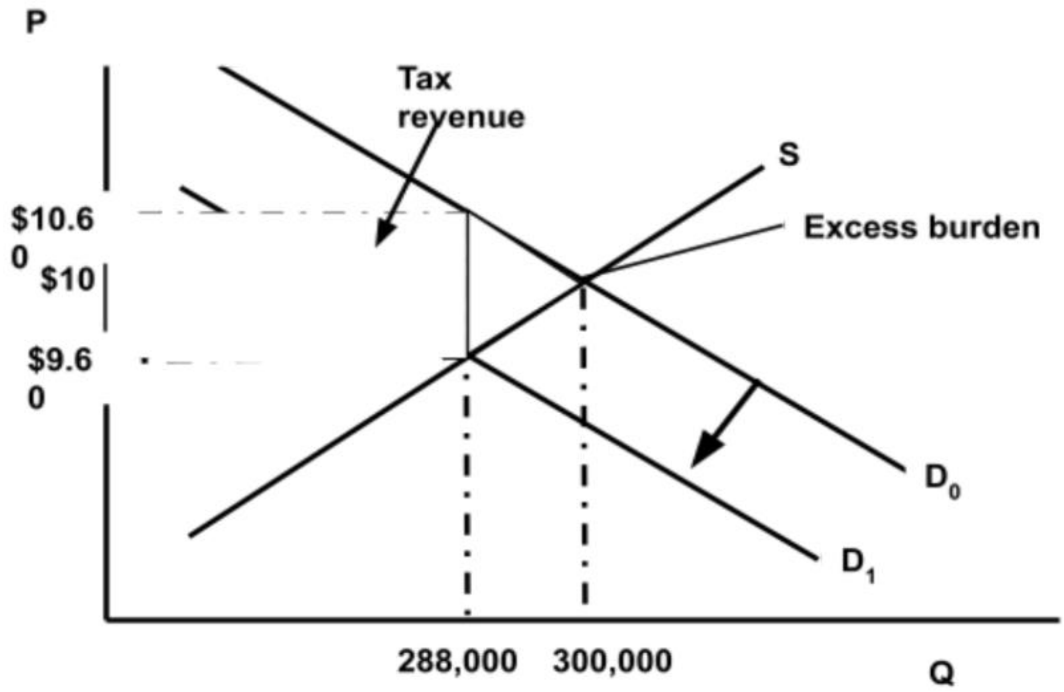
ANS

a. Before-tax equilibrium: P = $10 and Q = 300,000

After-tax equilibrium: P = $10.60 and Q = 288,000

Consumers pay $10.60 and producers receive $9.60.

Excess Burden = ½(12,000)($0.60) + ½(12,000)($0.40) = $6,000.



b. If the negative external cost were equal to $1 per gallon, then a $1 tax would achieve an efficient allocation and would create no excess burden. With a negative external cost of $0.50 per gallon, there is still an excess burden associated with a $1 per gallon tax, but it is smaller since the efficient level of output in this market would be between 288,000 and 300,000.